Fake News Detection using NLP & Explainable AI

Abstract

This project presents an AI-based system for detecting fake news using Natural Language Processing (NLP) and Machine Learning techniques. The model uses TF-IDF for feature extraction and Logistic Regression for classification. Additionally, SHAP is integrated to provide interpretability of model predictions, making the system more transparent and reliable.

Introduction

In today's digital world, the spread of misinformation has become a major concern. Fake news can influence public opinion and create confusion. This project aims to develop an automated system that analyzes news content and classifies it as real or fake using machine learning techniques.

Tools Used

- Python
- Streamlit
- Scikit-learn
- NLTK, Pandas
- NumPy
- Matplotlib
- SHAP

Steps Involved in Building the Project

1. Data Collection: Real and fake news datasets were used.
2. Data Preprocessing: Text cleaning and stopword removal.
3. Feature Extraction: TF-IDF vectorization.
4. Model Training: Logistic Regression model.
5. Evaluation: Accuracy-based testing.
6. Deployment: Streamlit web app.
7. Explainability: SHAP integration.

Conclusion

The project successfully demonstrates fake news detection using machine learning. Explainable AI improves trust in predictions. Future improvements include deep learning models and real-time data integration.